CALIFORNIA ACADEMY OF SCIENCES

(http://californiaacademyofsciences.applytojob.com/apply)

Biodiversity Data Specialist

San Francisco, CA

Temporary

IBSS

Experienced

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Come join a team dedicated to the Academy's mission, vision *and values!* (https://www.calacademy.org/vision-values) Currently, the Academy has a new strategic plan (https://www.calacademy.org/our-strategic-plan)including three initiatives – *Hope for Reefs*, *Thriving California*, and *Resilient Islands* – that leverage biodiversity science, environmental learning, and collaborative engagement to regenerate fragile ecosystems around the world. Learn more at https://www.calacademy.org/about-us/major-initiatives (https://www.calacademy.org/about-us/major-initiatives).

We hope you are inspired by what we do and are excited to contribute to our mission. The Academy is looking for candidates who do great work, and we know they may come from a number of different backgrounds and experiences. We encourage you to apply even if you do not believe you meet every one of the qualifications for the position.

Please note: As of Friday, August 27, Academy employees and volunteers are required to provide proof of COVID-19 vaccination or undergo testing and provide negative test results before they are able to work on-site or off-site with others.

POSITION SUMMARY:

The Biodiversity Data Analyst/ Scientist designs innovative approaches to integrate, analyze, model, synthesize, and visualize community-contributed biodiversity observations arising from the activities of the Center for Biodiversity and Community Science and

the iNaturalist community, with the ultimate goal to reveal patterns of biodiversity change and produce actionable biodiversity insights in California and globally. The Biodiversity Data Analyst/ Scientist is in charge of extracting important biodiversity information held in community-contributed data and museum collections and translating it into insights relevant to research and management of natural resources. In addition, the Biodiversity Data Analyst/ Scientist leads the development of innovative tools to mobilize, analyze, and visualize biodiversity data from community science and natural history collections within Academy-wide initiatives aimed at informing critical science, conservation, and decision making. This position is grant-funded for two years with a possibility for extension.

The Center for Biodiversity and Community Science uses the biodiversity observation recording and social network platform iNaturalist (https://www.inaturalist.org) for the bulk of our data collection. The Biodiversity Data Analyst/ Scientist will lead the development of data analyses that integrate community science collected and museum specimen species occurrence (presence only) data with available environmental data (temperature, salinity, pH, etc), to build predictive spatial models for species. In addition to providing overall expertise in working with large biodiversity data sets (primarily species occurrence data) across the organization, the Biodiversity Data Analyst/ Scientist will work on three specific projects:

1. *Early Warning and Forecasting System for Biodiversity Change on the California Coast*: Building on preliminary work, the Postdoctoral Researcher will use community collected species occurrence data to complete a tool that forecasts changes in species abundance based on climate projections. 40% time
2. *Bringing Urban Biodiversity Data and Social Equity to Conservation Planning in the Bay Area*: The researcher will build species distribution models using community collected data for 10 Bay Area species and be a key member of a team incorporating species occurrence data into a regional planning tool. 20% time
3. *Visualizations for Academy Strategic Initiatives*: The researcher will build interactive visualizations, as needed, using collections and community science data for the Thriving

California (https://www.calacademy.org/about-us/major-initiatives/thriving-california) and the Islands 2030 (https://www.calacademy.org/about-us/major-initiatives/resilient-islands) initiatives. 25% time

4. *Other projects as needed* 15% time

The Center for Biodiversity and Community Science is within the Academy's Institute for Biodiversity Science & Sustainability. We are committed to developing a community of naturalists working together to document biodiversity, connecting people to their local nature, and simultaneously collecting data critical to science and management. We accomplish these goals through a variety of programs including community bioblitzes, long-term intertidal monitoring, City Nature Challenge (https://citynaturechallenge.org/), and Snapshot Cal Coast (https://www.calacademy.org/calcoast), which scaffold collecting biodiversity observations using the iNaturalist platform with in-person community science field experiences. Our team is also dedicated to building best practices around engagement through community science and is actively working on research related to science learning and development of environmental science agency through participation in community science. The Center for Biodiversity and Community Science has delivered a broad range of innovative projects since 2012, reaching tens of thousands of people and collecting millions of species occurrence records throughout California and beyond.

We are committed to ensuring diversity, equity, accessibility, and inclusion are intrinsic to Academy culture and operations, from recruitment and retention to science and advocacy. Explore Our Commitment to Racial Equity (https://www.calacademy.org/our-commitment-to-racial-equity) *to learn about our intentional engagement of traditionally excluded groups in an effort to spark interest in or foster pathways to scientific careers.*

POSITION DUTIES AND RESPONSIBILITIES :

- Integrates, analyzes, models, and visualizes community-contributed biodiversity observations with additional data sources to answer key research and management questions and indicators useful for decision making
- Determines computational strategies for investigating scientific questions concerning the ecology and distribution of species and ecological communities, as well as the integration of community science efforts into land and coastal management

- Contributes to Academy initiatives by visualizing species occurrence data, developing other helpful visualizations.
- Designs and builds software and web-based interactive tools to visualize biodiversity insights derived from community-contributed and additional data
- Builds models and tools to forecast biodiversity responses to likely future environmental changes in California
- Provides regular updates on project progress to funders and partners
- Disseminates findings in written technical reports and academic publications, as well as oral presentations at scientific conferences and public meetings
- Contributes to content development of grants to support activities of the Center for Biodiversity and Community Science related to analysis and modeling of biodiversity data
- Participates in the Academy's fieldwork to become familiar with community science data collection and how to best use it for modeling purposes
- Travels to Sacramento and throughout the Bay Area to meet with project funders, stakeholders, and partners, as needed.
- Works early mornings, evenings, and weekends, as needed
- Follows all Academy safety regulations
- Other duties as assigned

QUALIFICATIONS:

EXPERIENCE and/or EDUCATION:

A qualified candidate will possess a combination of the following experience and/or equivalent education:

- Researchexperience and expertise in spatial ecology, species distribution modeling, data visualization, mapping, museum collections data, etc., and experience and/or interest in both advancing scientific frontiers and addressing real-world challenges in sustainable ocean resource management.
- Experience in geo-spatial analysis and data visualization (R, QGIS, ArcGIS, etc.),
- Experience in building web-based interactive mapping and dashboard applications in RShiny
- Experience in reproducible workflows, code version control, and management (Git, GitHub)
- Ph.D. degree in evolutionary biology, ecology, statistics,

bioinformatics, data science, or related fields.

KNOWLEDGE, SKILLS, AND ABILITIES:

- Expertise in modeling past and future impacts of environmental change on the distribution of species and ecological communities
- Expertise in modeling spatial and temporal variation in species, functional and phylogenetic diversity of ecological communities
- Expertise in R programming and statistical environment for the integration, manipulation, and analysis of big data
- Proficiency with harvesting data from the web using Application Programming Interfaces (APIs) and web scraping
- Excellent verbal and written communication skills (English) and a strong background in science communication with diverse audiences
- Strong interest in biodiversity and ability to communicate that passion
- Works as a part of multi-disciplinary teams and diverse stakeholder groups
- Committed to working on a team that is striving to be anti-racist in work, interactions, systems, and tools
- Confident and flexible communicator, both written and verbal, to diverse audiences
- Excellent organizational skills and attention to detail
- Ability to work independently, and as a member of interdisciplinary teams on cross-cutting issues
- Demonstrated ability to successfully collaborate with non-scientists
- Flexible and creative
- Ability to read and interpret documents such as safety rules, operating and maintenance instructions, and procedure manuals

WORK ENVIRONMENT:

Reasonable accommodations will be made to enable individuals with disabilities to meet the job's essential functions. While performing the duties of this job, the employee is frequently required to stand, sit, walk, use a computer, and communicate on the phone, in person, and online meetings/calls. The employee will spend time out-of-doors at events and programs, occasionally in

hot or inclement weather. Must be able to transport 30 lbs. Must be able to perform repetitive motions 50% of the time.

APPLICATION DEADLINE:

January 7, 2022, 5pm. Applications will be reviewed on a rolling basis until the position is filled.

APPLICATION PROCESS:

Please upload your resume and complete the brief online application.

The California Academy of Sciences will give full consideration for employment to all qualified applicants with criminal histories in a manner consistent with the requirements of the San Francisco Fair Chance Ordinance (https://sfgov.org/olse/fair-chance-ordinance-fco) **(SF Police Code, Article 49).**

The California Academy of Sciences is an Equal Opportunity Employer and is committed to ensure that all employees and applicants receive equal consideration and treatment, regardless of race, color, creed, gender (including gender identity or gender expression), religion, marital or domestic partner status, age, place of birth, national origin or ancestry, physical, mental or medical disability, height or weight, sex, sexual orientation, citizenship, military service status, veteran status, or any other characteristic protected by state or federal law or local ordinance.

Apply for this position

REQUIRED *

First Name *

Last Name *

Email Address *

Phone *

Address *

Address

City

State/Province

Postal

Resume *

Attach resume or Paste resume

Cover Letter *

LinkedIn Profile URL:

Desired salary

Earliest start date?

Preferred Name

Pronouns

Have you worked at the Academy before? *

-- No answer --

If yes, when and what was your job title?

How did you hear about this position? If you were referred, please list name below. *

What, if any, accommodations would you need if selected for an interview?

Briefly describe your experience integrating different types of data and/or developing species distribution models to answer scientific or management questions. *

Briefly describe your experience in producing and making data visualizations and maps to communicate with academic and non-academic audiences. *

In a few sentences, what role or impact do you think species occurrence data, especially community science and museum collections data, can have in research, conservation, and management? *

Briefly describe your experience in science communication (print and/or oral) to diverse audiences. If you have one available, please provide a link to an example of you communicating your science to a non-academic audience. *

Please briefly describe any experience you have working on questions related to California biodiversity or if you don't have experience working in California, please describe any place-based work you have done. What's your favorite California species (or species from the place you detailed above) and why? *

The following questions are entirely optional.

To comply with government Equal Employment Opportunity and/or Affirmative Action reporting regulations, we are requesting (but NOT requiring) that you enter this personal data. This information will not be used in connection with any employment decisions, and will be used solely as permitted by state and federal law. Your voluntary cooperation would be appreciated. Learn more (https://www.eeoc.gov/employees-job-applicants).

Gender

Decline to answer

Race/Ethnicity

Decline to answer

SUBMIT APPLICATION

POWERED BY JAZZHR (HTTPS://WWW.JAZZHR.COM)